

Leveraging Interpolation Models and Error Bounds for Verifiable Scientific Machine Learning

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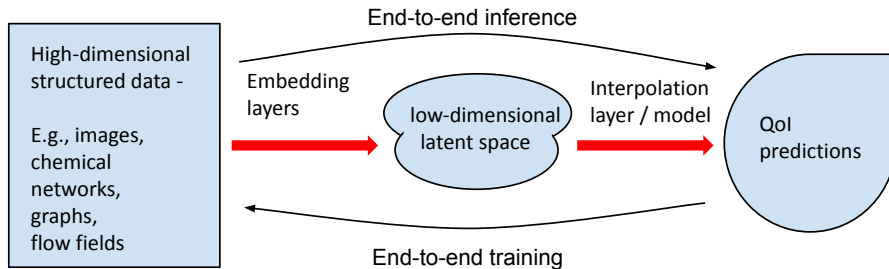
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AI, SciML, and Interpolation

$AI = Representation\ Learning + Regression/Interpolation$



SciML = **separate** representation learning and regression or interpolation

This paper = **just interpolation** in the context of SciML

Outlines

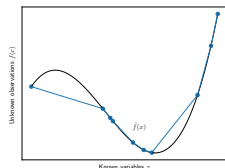
Interpolation Models and Error Bounds

Experiments on Synthetic Data

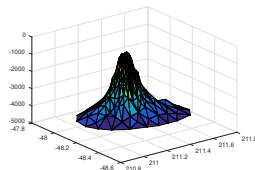
Experiments on Airfoil SciML Problem

Piecewise Linear (Delaunay) Interpolation

Approximate $f(q)$ by piecewise linear $\hat{f}_T(q)$; $T =$ simplicial mesh



1 dimension



d dimensions

If $S \in T$ contain q ; $\text{diameter}(S) = h$:

$$|f(q) - \hat{f}_T(q)| \leq \left(\frac{\gamma + \gamma\sigma\sqrt{d}}{2} \right) h^2$$

$\gamma =$ Lip const of ∇f ; $\sigma =$ condition number of interpolation matrix (skew of S);

Gaussian process interpolation

Two interpretations:

1. $\hat{f}_{GP}$ = RBF interpolant with “Gaussian bump” kernel
2. $\hat{f}_{GP}$ = posterior distribution over all possible realizations of f given the observations in the training data

(2) lets us use the variance of q as an uncertainty heuristic:

$$K(q, q) = \tau_f \left(1 - \|\phi^{(1)}(q), \dots, \phi^{(1)}(q)\|_{A_{GP}^{-1}}^2 \right)$$

$\phi^{(1)}, \dots, \phi^{(n)}$ are Gaussian basis functions; $\tau_f \sim$ variance of f ; $\|\cdot\|_{A_{GP}^{-1}}$ is a rescaling of $\|\cdot\|_2$

Neural Network models (MLPs)

In a typical modern AI model:

$$\hat{f}_{ANN}(Q) = \hat{f}_{MLP} \circ \hat{f}_R(Q)$$

where

$$\hat{f}_R(Q) : ? \rightarrow \mathbb{R}^d$$

and

$$\hat{f}_{MLP}(q) = \ell^{(N)} \circ \ell^{(N-1)} \circ \dots \circ \ell^{(0)}(q)$$

Each

$$\ell^{(i)}(x^{(i)}) = u\left(W^{(i)}x^{(i)} + b^{(i)}\right)$$

where $W^{(i)}$ and $b^{(i)}$ define a linear transformation, and u is a nonlinear “activation function”

We will focus on MLPs which use ReLU (piecewise linear) activation function which is the most common option

Experiments on synthetic datasets

For all the interpolation models we saw constants that depend on:

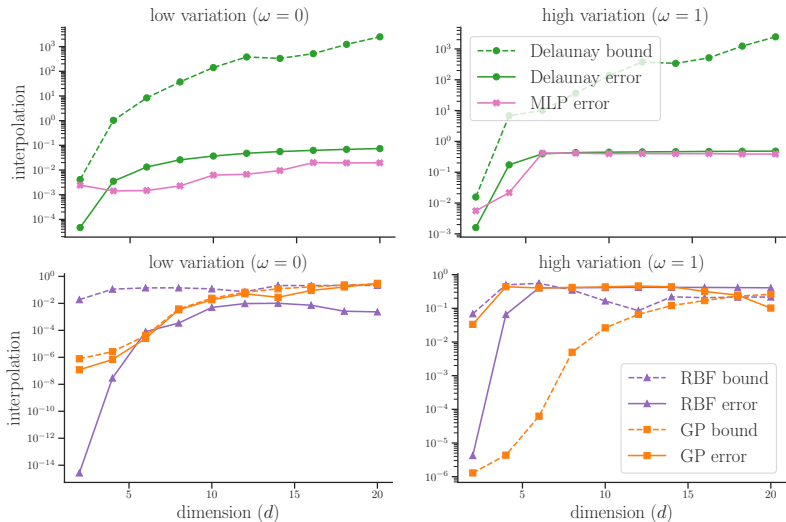
1. the dimension of the problem
2. the density of the training data,
3. the smoothness of f ,
4. the conditioning of the interpolation matrices,

We generate hundreds of datasets from different distributions varying the

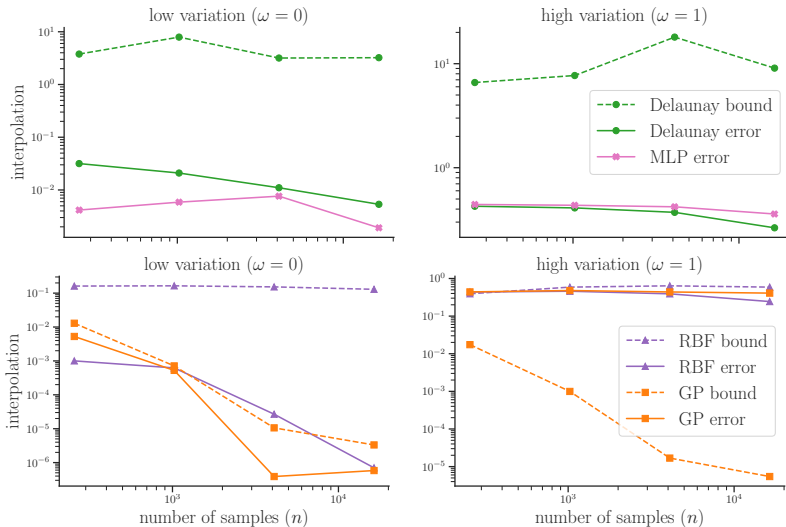
1. dimension d
2. number of training points n
3. Lipschitz constant of the underlying f
4. skewness of the training points

We use all methods on all training sets and measure the prediction errors on a holdout set generated from the same distribution.

Effects of problem dimension and smoothness



Effects of training sample size and problem smoothness



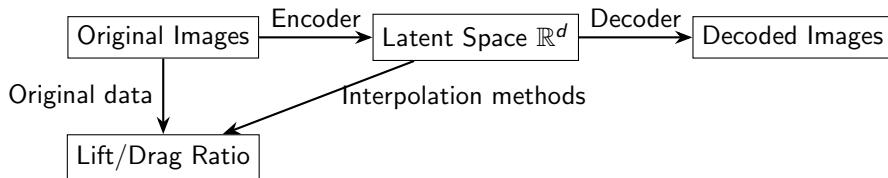
Experiments on UIUC Airfoil Dataset

Training set:

~1600 labeled images: $f(\text{image}) \rightarrow \text{lift/drag}$

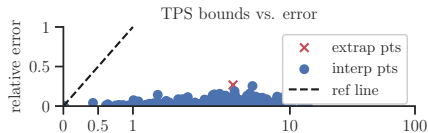
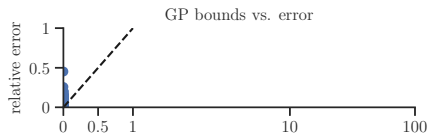
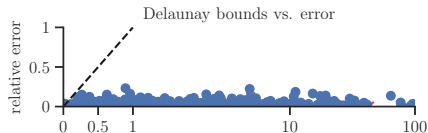
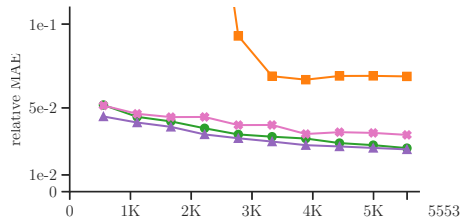
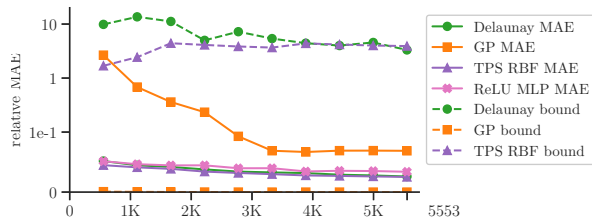


Problem representation:



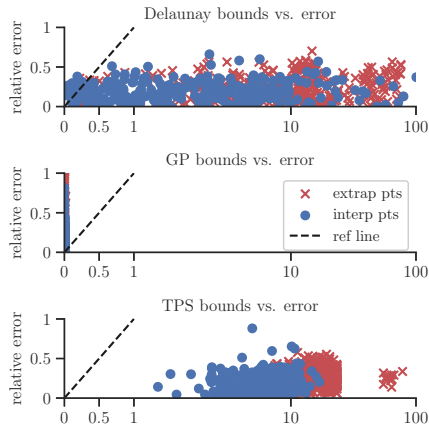
<https://github.com/ziliHarvey/CNN-for-Airfoil>

Results on Validation Set

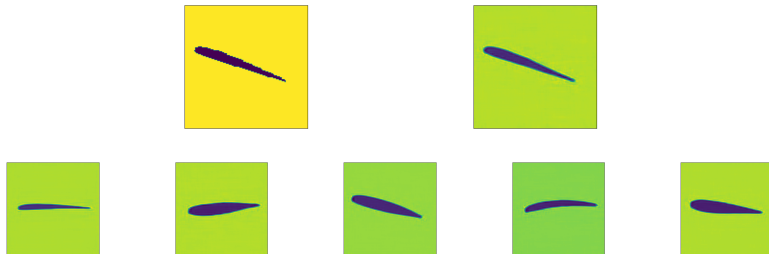


Results on Test (Holdout) Set

Method	MAE	average bound
Delaunay	0.21	20.04
GP	17.38	0.01
TPS RBF	0.23	12.87
ReLU MLP	0.21	n/a



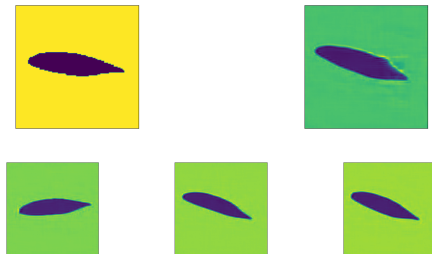
Diving Deeper on the Validation Set



Top: True validation image (left) and its embedding (right).

Bottom: The 5 training images used to make the prediction.

Diving Deeper on the Test Set



Top: True test image (left) and its embedding (right).
Bottom: The 3 training images used to make the prediction.

Conclusions

- ▶ Error bounds for interpolation methods are useful diagnostic tools for understanding performance of these methods
- ▶ Interpolation methods and their error bounds can be used to detect potential issues at various stages of scientific machine learning training/validation/testing pipeline
- ▶ The right interpolation methods achieve similar performance to deep learning methods on real-world noisy *regression* problems
- ▶ Interpolation methods should be a tool for scientific machine learning problems, alongside other AI-based approaches

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Questions

Interpolation Models and Error Bounds

Experiments on Synthetic Data

Experiments on Airfoil SciML Problem



github.com/LLNL/interpML

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